

period is 2 ms (2000 $\mu$ s). The output pulse width varies from 500  $\mu$ s to 1500  $\mu$ s. The R-C time constant of the monostable multivibrator is kept slightly less than 500  $\mu$ s to generate exactly the same inverted PWM as is generated by the microcontroller.

When switch S2 is pressed, port-pin P2.5 goes high and RL1 de-energises to stop the motor.

When switch S3 is pressed, relay RL2 energises. Pin IN1 of motor driver L293D receives the PWM signal and pin IN2 connects to ground. As a result, the motor rotates in one direction (say, clockwise).

When switch S3 is pressed again, relay RL2 de-energises. Pin IN2 of motor driver L293D receives the PWM signal and pin IN1 connects to ground. The motor now rotates in opposite direction (anti-clockwise).

When switch S4 is pressed, different modes are selected in cyclic manner as given below:

1. *Continuous mode.* The motor rotates continuously with the set speed in either direction

2. *Reversible mode.* The motor reverses automatically after the set time

3. *Jogging mode.* The motor rotates for the set time in either direction and then stops for a few seconds and again rotates for the set time. It is also called 'pulse rotation'

Switches S5 and S6 are used to set the speed of the motor, either in increasing order or decreasing order, in continuous mode only.

Switches S7 and S8 are used to set the time either in increasing order or decreasing order.

## Software

The program is written in 'C' language and compiled using Keil  $\mu$ Vision3 compiler. It is well commented and easy to understand. The program has three major sections: initialisation and key press, display and motor control.

**Initialisation and key press.** It consists of the main function that initialises timers, ports, LCD module and LED indication and then waits for key press.

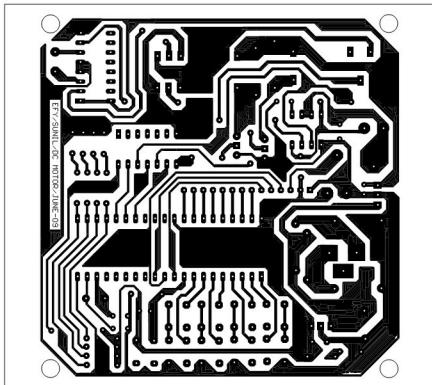


Fig. 4: A single-side, actual-size PCB layout for the microcontroller-based DC motor controller

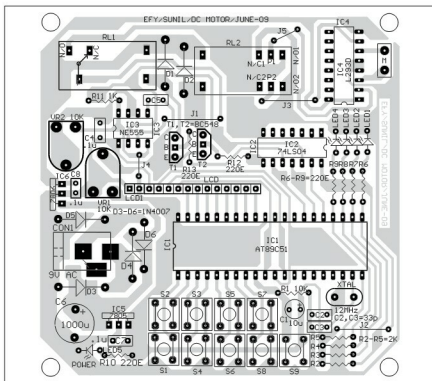


Fig. 5: Component layout for the PCB

When any key is pressed, the program calls that particular function.

In the main function, first the ports are initialised as input/output, then